

ClawFarm: A Protocol for Mining AI Inference

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Abstract

ClawFarm is a protocol for mining AI inference. Providers register endpoints, users deposit USDC into non-custodial escrow, and each settled call produces a dual-signed proof that determines payment and CLAF emission. Providers select a protocol-fee tier between 0.5 percent and 3.0 percent, in 0.5 percent increments. For settlement amount A and selected rate r , treasury receives $A \times r$ and provider pending revenue receives $A \times (1 - r)$. CLAF emission follows a ten-year halving schedule, with 70 percent assigned to providers and 30 percent assigned to developers and users, weighted by actual protocol-fee contribution. The target treasury policy routes protocol fees through predetermined buyback, maintenance, and infrastructure paths. The protocol does not verify model identity or inspect inference content. It is source-blind by design, allowing any wallet, endpoint, and inference source to enter the same settleable network.

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1. Purpose

ClawFarm is a protocol for mining AI inference. It treats inference as an economic output that can be metered, priced, settled, and rewarded without requiring a platform operator to decide who may supply it or who may consume it.

The central risk of the AGI era is not that models become more capable. It is that model capability, compute access, settlement rails, and reward distribution become locked inside a small number of institutional balance sheets. If the supply side of machine intelligence is captured, every application above it becomes downstream of the same narrow choke point.

ClawFarm is designed to separate the supply side of machine intelligence from any single company, marketplace, or account system. Any wallet, any endpoint, and any source of inference capacity can enter the same settleable network, subject only to the protocol's proof, bond, settlement, and challenge rules.

2. What the protocol is

ClawFarm is not a model lab, a cloud provider, an inference reseller, or a hosted application. It is a settlement protocol for inference calls. The protocol records provider registration, escrowed user funds, dual-signed usage proofs, settlement events, treasury inflows, treasury split events, mining rewards, burn events, and protocol-owned liquidity additions.

Applications use a chat-completion compatible interface. Providers register an endpoint and declare model, price, and quality information. A call settles only after the user and provider both sign the same usage proof. The protocol does not inspect the inference itself and does not claim to verify model identity.

This narrowness is deliberate. A protocol that decides which provider is legitimate becomes a platform. A protocol that only settles signed usage can remain identity-blind, source-blind, and forkable.

3. Supply neutrality

Inference capacity is not a single kind of resource. It may come from closed-model API resale, subscription credit pools, self-hosted open-weight models, leased GPU capacity, colocated infrastructure, or future sources that do not exist yet. The protocol does not record which source a provider uses.

This is a structural property, not a slogan. Registration asks for a wallet, a bond, an endpoint, and declared offerings. Settlement asks for a dual-signed proof. Reward accounting reads settled volume and price. None of these operations requires the protocol to ask where capacity came from.

The treasury operates with the same neutrality. Buyback and add-LP slices are submitted by any wallet that pays gas; the program enforces the bounds, not the identity of the caller. Maintenance and infrastructure withdrawals can only flow to addresses declared at Genesis and have no power over the buyback engine.

Supply neutrality is the mechanism by which AGI capacity can be opened at scale. The protocol cannot make compute abundant by itself. It can remove the settlement and reward bottleneck that would otherwise force capacity through a small number of intermediaries.

4. Roles

A developer or user deposits USDC into an escrow PDA and consumes inference through an application or SDK. Funds leave escrow only through withdrawal or settlement against a dual-signed proof.

A provider registers an endpoint by posting a 100 USDC bond. The provider selects a protocol-fee tier from 0.5 percent to 3.0 percent, in 0.5 percent increments. The provider receives the settlement remainder after the selected fee and earns CLAF from the Provider Pool according to actual fee contribution, price weight, and quality factors.

A challenger is any party that posts a challenge bond against a suspect settlement. Challenges are permissionless. Enforcement is therefore not operator-driven; it is carried by the economic incentives of participants who can prove a fault.

5. Escrow

User funds are held in a Program Derived Address controlled by the escrow program. No private key controls the escrow account. No administrator can sweep balances. The program releases funds only according to the rules deployed at Genesis.

A user may withdraw available funds at any time, subject to unsettled obligations. A settlement instruction transfers the provider share to the provider wallet and the protocol fee to the treasury PDA. The settlement path is deterministic and does not require human approval.

6. Dual-signed proof

A usage proof binds the request hash, response hash, model identifier, input token count, output token count, agreed price, timestamp, user wallet, and provider wallet. The user signs the proof. The provider signs the same proof. Settlement requires both signatures.

Neither side can unilaterally distort the settlement amount. A provider cannot overstate tokens without the user's signature. A user cannot understate tokens without the provider's signature. The protocol validates signatures and accounting, not the semantic quality of the answer.

This distinction matters. The protocol does not know whether an answer is good, whether a model is the exact version declared, or whether the upstream source is a frontier lab, a subscription account, or a local GPU. It knows that both parties signed the same metered receipt.

7. Settlement

For a settled call, the payment amount A is calculated from the provider's declared price and the metered usage in the signed proof. The provider selects a protocol-fee rate r from the allowed set: 0.5 percent, 1.0 percent, 1.5 percent, 2.0 percent, 2.5 percent, or 3.0 percent. Treasury receives $A \times r$. Provider pending revenue receives $A \times (1 - r)$.

The fee tier is provider-selected, not governance-selected. Lower fee tiers reduce the treasury contribution and reduce CLAF reward weight in the same proportion. A provider may choose lower friction for users, but cannot keep the same mining weight while contributing less to the protocol.

The protocol-fee inflow is split on-chain into three predetermined vaults at fixed ratios: 70 percent buyback, 20 percent maintenance, and 10 percent infrastructure. No discretionary allocation to a foundation, contributors, or marketing exists. The split, the recipient addresses, and the withdrawal paths are all set at Genesis and cannot be redirected.

8. Mining and emission

Each settled inference call mines CLAF. Emission is divided between two pools: 70 percent to the providing side and 30 percent to the consuming side. The schedule runs for ten years and halves every two years.

The maximum emitted supply over the schedule is approximately 968.75 million CLAF. The remaining approximately 31.25 million CLAF is never emitted by the protocol. Total supply is fixed at 1,000,000,000 CLAF. No portion of the schedule is pre-allocated to any party. The team participates in mining only as a settling consumer or provider, on the same terms as any other wallet. Where team-mined CLAF is used for initial pool seeding in Appendix A, it is surrendered to the pool and the resulting LP is burned, ensuring the team retains no claim to that liquidity.

Within each pool, reward weight follows actual protocol-fee contribution rather than nominal settlement volume alone. A call settled at a lower fee tier contributes less mining weight than the same call settled at a higher fee tier. Token distribution therefore follows the amount of USDC fee paid into the protocol.

Provider rewards are additionally weighted by price relative to the network average. A provider that clears below the network average price receives a higher CLAF weight for the same fee contribution. The mechanism subsidizes production of inference when USDC clearing price is below immediate marginal cost.

Developer rewards are earned by settled consumption and inherit the same fee-contribution weighting. This gives the demand side a share of emission and helps bootstrap both sides of the network at the same time.

9. Treasury and burn

The protocol treasury receives the provider-selected fee from every settlement in USDC. The allowed fee tiers range from 0.5 percent to 3.0 percent in 0.5 percent increments.

Each epoch, accumulated USDC is split deterministically: 70 percent to a buyback vault, 20 percent to a maintenance vault, and 10 percent to an infrastructure vault. The split is enforced on-chain and the split crank is permissionless.

The buyback vault drives two non-discretionary actions against the CLAF/USDC liquidity pool: buy-and-burn and buy-and-add-LP. Buy-and-burn removes CLAF from circulation. Buy-and-add-LP buys CLAF with treasury USDC, then deposits the bought CLAF plus an equal share of treasury USDC into the pool; the minted LP tokens are sent to a code-locked vault from which the protocol has no withdrawal path. The split between the two actions is a function of the pool's liquidity ratio $L = \text{pool TVL} / \text{circulating market cap}$, with a target band that prefers add-LP when L is below target and burn when L is above target.

Slices within a treasury execution window are randomized in count, size, and timing within bounds enforced on-chain: min and max slice size, min interval, cumulative cap, and min-out slippage tolerance. Randomization is sampled from a CSPRNG and submitted through bundled execution where available, to limit predictability and frontrunning. The on-chain swap is a direct CPI to a constant-product pool program; the swap instruction is composed off-chain so a different pool program can be substituted without contract changes.

The maintenance vault and infrastructure vault fund the operational and infrastructure costs that keep settlement, indexing, and challenge handling available. Both vaults pay out only to addresses fixed at Genesis and recorded on-chain.

No human trigger initiates a buyback. An epoch is opened by anyone; slices are executed by anyone; the split crank is called by anyone. The maintenance and infrastructure vaults are the only paths where an authorized address moves funds out of the treasury, and those addresses can withdraw only to themselves.

9a. Anti-MEV

Two layers protect buyback execution from frontrunning and price manipulation: per-slice min-out and randomized timing and size.

Every swap instruction enforces a minimum CLAF output, computed off-chain from the current pool reserves using the same constant-product math the on-chain pool program runs. The maximum slippage tolerance is a configurable parameter bounded by the admin. A swap that would receive less than the min-out reverts on-chain.

Within each treasury execution window, slice count, individual size, and firing times are drawn from a CSPRNG and constrained by on-chain bounds: min and max slice size, min interval, and cumulative cap. Adjacent slices respect the min interval. Where bundled execution is available, slices are submitted in private bundles to limit mempool visibility.

A minimum pool liquidity threshold prevents buyback during periods of thin or drained liquidity. The threshold is admin-tunable and starts at a conservative floor relative to the seed pool size.

The protocol does not consult an external price oracle. Adding an on-chain oracle as a third defense layer was considered and deferred: at launch the pool is shallow enough that a pool-derived TWAP is itself cheaply manipulable, and CLAF has no independent oracle feed. The composition of min-out, randomized execution, and liquidity floor is the operating defense; an external oracle may be added later if it becomes practical.

10. Registry and routing

The registry stores provider wallets, endpoints, declared models, pricing, and status. It does not store legal identity, capacity source, or upstream account information. Any wallet that posts the required bond can register.

Routing is performed by clients and applications against registry data. A route may prioritize price, latency, quality history, or a custom policy selected by the developer. The protocol does not operate a central router.

The registry makes supply legible without making it permissioned. Developers can see which models have independent wallet-backed supply and what prices are being cleared, while providers remain peers rather than approved vendors.

11. Bond and challenge

Provider registration requires a 100 USDC bond. The bond creates a cost to spam registration and gives the challenge system an economic anchor. Deregistration includes a cooldown so recent settlements can still be challenged.

Any party may challenge a suspect settlement by posting a small bond. If the challenge is upheld, the provider is slashed and the challenger receives a reward. If the challenge fails, the challenger forfeits the bond. The mechanism discourages both provider abuse and frivolous challenges.

Challenge-driven enforcement is weaker than perfect cryptographic verification of inference, but it is available now and preserves source neutrality. Future cryptographic proof systems can be layered above or forked into a new protocol if they become practical.

12. Governance scope

The protocol distinguishes between structural and operational state. Structural state - mints, vault addresses, fee split ratios, the recipient addresses for maintenance and infrastructure, the PDA seeds, and the program ID - is fixed at Genesis and cannot be changed by any party after deployment.

Operational state - epoch timing, slice size bounds, slippage tolerance, minimum pool liquidity threshold, and minimum buyback threshold - is bounded and can be tuned by an admin multisig. The admin has no authority over fund flows beyond withdrawing the maintenance and infrastructure vault balances to their pre-declared recipients. The admin has no authority over the buyback vault, the CLAF holding vault, or the protocol-owned liquidity vault.

Two multisigs hold authority at Genesis. An upgrade multisig holds program upgrade authority; all upgrades pass through an on-chain timelock with a 48-hour delay before they can execute. An admin multisig holds the bounded operational role above and also serves as the recipient address for the maintenance and infrastructure vaults; admin actions that affect fund flow pass through a 24-hour timelock. Both timelock durations and both multisig memberships are publicly visible on-chain.

An emergency pause exists. It is set and lifted by the admin multisig in its pauser role and is exempt from timelock to preserve its emergency function. Pause halts the buyback engine only; it cannot redirect, withdraw, or otherwise move funds. The pause role is renounceable: a single transaction permanently removes pause capability from the protocol.

The protocol does not commit to renouncing upgrade authority. A settlement protocol that custodies user USDC and operates against external DEX liquidity must retain the ability to patch safety-critical bugs. Renouncement would convert any future bug into a permanent loss for users. The trust model is bounded by structural immutability, on-chain timelocks, public multisig execution, and acknowledged launch-period signer trust assumptions.

During the launch period, signers on both multisigs are members of the founding team. This is acknowledged as a trust assumption: signer diversity expands over time, but the protocol is not fully trustless on day one and does not claim to be.

The protocol has no team allocation in the emission schedule and no investor allocation. Rewards follow settled contribution. A separate cold-start commitment describes how initial pool liquidity is provided by team-mined CLAF and immediately surrendered.

13. Security limits

ClawFarm does not verify model identity. It does not know whether a provider served the exact model declared. It does not know the provider's upstream source. It does not solve all collusion or wash-usage attacks. It imposes costs, signatures, bonds, and challenge incentives.

The protocol depends on Solana for execution, finality, censorship resistance, and token-account semantics. A base-layer failure or halt affects ClawFarm directly.

These limits are not hidden. They define the boundary of the design. ClawFarm is a settlement and mining protocol for inference usage, not a universal truth machine for artificial intelligence.

14. Cold-start commitment

CLAF launches with no pre-mined allocation and no pre-existing market. The protocol bootstraps initial liquidity by mining 10,000,000 CLAF, 1.0 percent of total supply, through ordinary settlement participation by the team's own wallets, on the same emission terms as any other participant. This portion is publicly verifiable on-chain.

The mined CLAF is paired with 5,000 USDC of team-provided capital, creating a Raydium CPMM pool at an initial fully diluted valuation of 500,000 USDC and an opening price of 0.0005 USDC per CLAF.

The LP tokens received from pool creation are burned. The team retains no claim on the seed liquidity. The 5,000 USDC and 10,000,000 CLAF become permanent pool depth that supports all subsequent buyback and add-LP activity.

Once revenue begins, treasury buyback adds further protocol-owned liquidity through the code-locked vault described in Section 9. Pool depth therefore grows with protocol usage, independent of any team or investor sales.

15. Conclusion

AGI capacity should not be locked behind a small number of accounts, balance sheets, and settlement channels. If intelligence becomes a core economic input, the rails that admit supply and distribute rewards matter as much as the models themselves.

ClawFarm makes inference supply permissionless and settleable. It lets any wallet register capacity, any application consume capacity, and both sides mine CLAF through real settled usage. The protocol does not ask who participates. It asks only that settled calls carry proof.

Compute may centralize. Accounts may centralize. Platforms may centralize. Settlement does not have to.

Appendix A. Genesis parameters

Parameter	Value
Chain	Solana
Settlement asset	USDC
Reward token	CLAF
Total supply	1,000,000,000 CLAF
Maximum scheduled emission	Approx. 968.75M CLAF
Unemitted residual	Approx. 31.25M CLAF
Emission horizon	10 years
Halving interval	2 years
Default epoch length	1 hour
Provider pool	70 percent of epoch emission
Developer pool	30 percent of epoch emission
Provider settlement share	99.5 to 97.0 percent of USDC settlement
Protocol fee	0.5 to 3.0 percent, provider-selected in 0.5 percent increments
Reward weight basis	Proportional to actual USDC protocol fee contributed
Treasury fee split	70 percent buyback / 20 percent maintenance / 10 percent infrastructure
Buyback action mix	Dynamic by liquidity ratio $L = \text{pool TVL} / \text{circulating market cap}$; target $L = 17.5$ percent. Cold start fixed at 70 percent add-LP / 30 percent burn until L exceeds target.
DEX	Raydium CPMM constant-product pool. Swap and deposit are composed off-chain and executed through treasury-controlled instructions.
Min buyback threshold	10 USDC at cold start; 100 USDC at steady state
Min slice size	1 USDC at cold start; 50 USDC at steady state
Max slice size	5,000 USDC
Min slice interval	30 seconds
Slippage tolerance	0.5 percent, configurable through bounded admin controls
Min pool liquidity	0 at deploy; raised to 5,000 USDC after pool seeding; intended to ratchet up
Protocol-owned liquidity	LP minted by buy-and-add-LP is sent to a code-locked vault with no withdrawal path
Initial pool seeding	5,000 USDC + 10,000,000 CLAF, with seed LP burned at pool creation
Provider bond	100 USDC mainnet target

Parameter	Value
Challenge mechanism	Permissionless bond and slash target
Upgrade authority	Upgrade multisig at Genesis, threshold 4-of-5, not renounced; upgrades pass through a 48-hour on-chain timelock
Admin / governance	Admin multisig at Genesis, threshold 3-of-5, bounded to operational parameters; fund-affecting actions pass through a 24-hour on-chain timelock
Emergency pause	Admin multisig pauser role, timelock-exempt, scoped to buyback engine, cannot move funds, renounceable
Maintenance recipient	Admin multisig; maintenance vault withdraws only to itself
Infrastructure recipient	Admin multisig; infrastructure vault withdraws only to itself
Total multisigs at Genesis	Two: upgrade and admin
Timelock durations	Upgrades 48 hours; admin fund-affecting actions 24 hours; pause exempt

References

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